CTOI 147035057.01 (TYC 5963-01913-1): three TESS eclipses fix the period at 37.315 d; the eclipsing body is most likely a star

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Summary

CTOI 147035057.01 was submitted in February 2021 (ExoFOP user *steuer*, "in-house single transit finder") as a single transit in Sector 33. It has no follow-up on ExoFOP. TESS shows two more eclipses of the same depth (~3.2 ppt) and shape (V-shaped, about 6.6 h from first to last contact):

• **Sector 7:** an earlier eclipse at BTJD 1515.065, in the TESS-SPOC and QLP light curves.

• **Sector 108:** a new eclipse at BTJD 4313.689 (2026-09-30). We found it with our own aperture photometry on the TESScut FFIs, because S108 has no SPOC/QLP light curve yet.

The three events fit a linear ephemeris (O–C ≤ 1.3 min, χ^2 = 0.97 for 1 dof), and every shorter alias is excluded by flat TESS data. Of the four aliases that predicted an S108 eclipse, only P = 37.315 d shows one. Difference images put the source on the target.

The period is fixed. The eclipsing body is uncertain, but most likely stellar.

• **The eclipse is grazing.** We fitted the eclipse with b and k = R_2/R_1 free and limb darkening held to physical values. The whole posterior is grazing: b = 1.28 (+0.27/–0.23), k = 0.34 (95%: 0.09–0.75). A V-shaped (grazing) model beats the best full-transit model by ΔBIC ≈ 10.

• **So the depth only gives a lower limit on size:** $R_2 > 1.7\ R_{Jup}$ (97.5%) for $R_1 = 2.0\ R_{sun}$. That is larger than expected for a planet or brown dwarf on a 37-d orbit.

• **Gaia RVs.** The host is strongly RV-variable: rv_chisq_pvalue = 0, rv_renormalised_gof = 99.8, rv_amplitude_robust = 49.8 km/s over 27 epochs. If K ≈ 25 km/s, the mass function is 0.060 M_{sun}, so $M_2 \geq 0.64\ M_{sun}$ (for e = 0). A K/M dwarf of about 0.6 R_{sun} gives k ≈ 0.3, consistent with the grazing fit. No secondary eclipse is seen, so the orbit is probably eccentric.

• **An open tension.** A companion like that should move Gaia's photocentre by about 0.5 mas. Yet RUWE is only 1.12. On the other hand, astrometric_excess_noise is 0.12 mas (28σ) and duplicated_source is set.

• **An alternative: two near-equal stars (SB2).** Such a pair would blur the RVs without moving the photocentre, and would dilute the eclipse. Gaia's spectroscopic log g (4.06) is closer to the single-star value (3.99) than to twins (4.23). That weakly favours one star. Either way, the eclipsing body is a star.

• **Gaia DR4 (2 Dec 2026)** should settle this: its epoch RVs, which include a double-lined product, can be phased on 37.315 d.

Ephemeris (BJD_TDB)	Value
P	37.31497 ± 0.00003 d
T0	2458515.0661 ± 0.0011 (BTJD 1515.0661)
Eclipse times (BTJD)	1515.0652 ± 0.0014 (cycle 0) ; 2224.0515 ± 0.0013 (cycle 19) ; 4313.6885 ± 0.0015 (cycle 75)
Depth / duration	3.19, 3.28 ppt (S7, S33); 2.8 ppt (S108, own photometry); first-to-last contact ≈ 6.6 h (grazing model; the catalogue box width is ~4.5 h)
Next eclipse well placed from Paranal	2027-01-20 03:11 UTC ± 3.4 min (alt 79°)

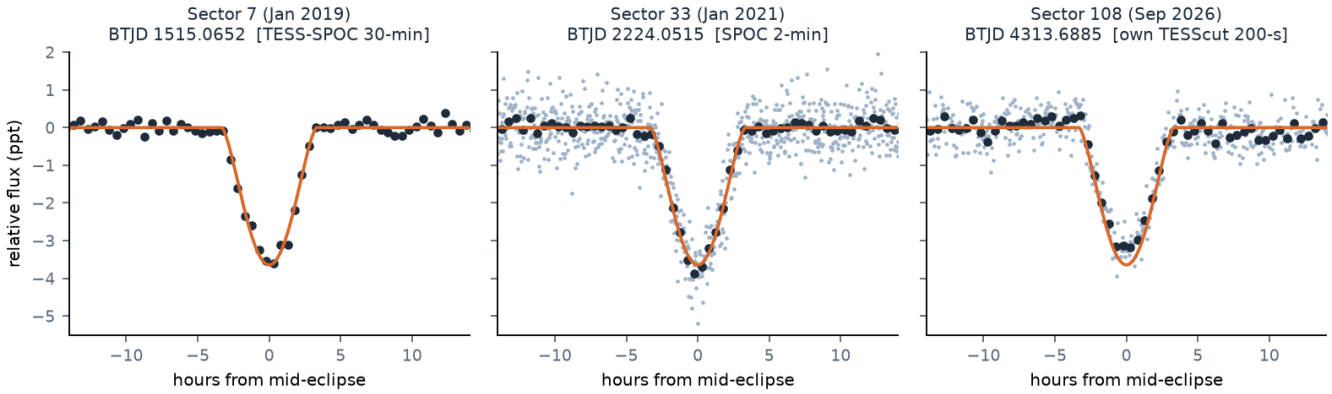


Figure 1. The three eclipses, detrended locally, with one common transit-shape model (orange). Grey: individual cadences; dark: 30-min bins (or all points for S7's 30-min data).

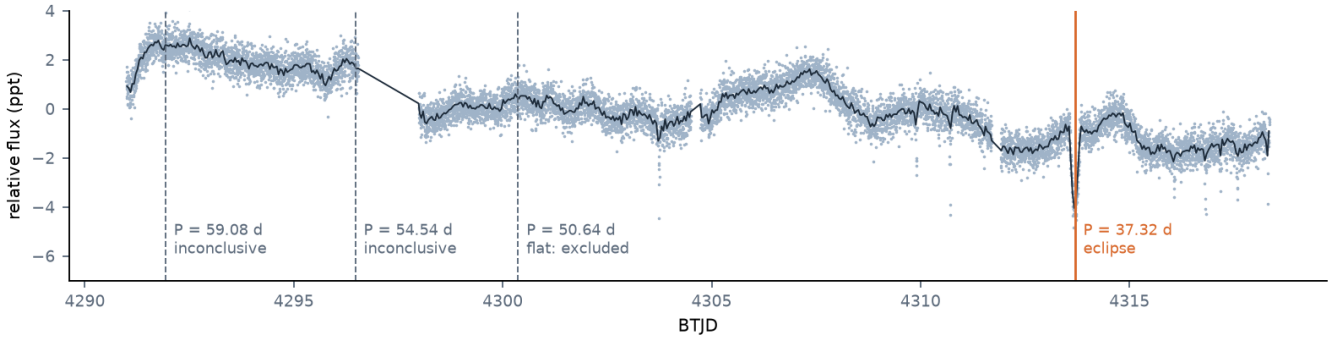


Figure 2. Sector 108 (own TESScut aperture photometry, 200 s). Vertical lines: predicted eclipse times for every alias still allowed before S108. Only P = 37.315 d shows an eclipse; P = 50.64 d falls on flat data (excluded), and P = 54.54 and 59.08 d fall on a gap edge and a ramp.

Star

Identifier / quantity	Value	Source
TIC / TYC / Gaia DR3	147035057 / 5963-01913-1 / 2947647054662200192	ExoFOP
RA, Dec (J2000)	104.27440, –16.47848	TIC
Tmag / G / BP–RP	8.49 / 8.83 / 0.62	TIC, Gaia DR3
Distance	166 pc (parallax 6.023 ± 0.017 mas)	Gaia DR3
Teff / log g	6400 K / 3.90 (GSP-Phot); 6480 K (TIC)	Gaia DR3, TIC
R* / M* / age	2.00 R_{sun} / 1.44 M_{sun} / 2.4 Gyr	Gaia DR3 FLAME
Astrometry	RUWE 1.12, ipd_frac_multi_peak 0, no NSS orbit	Gaia DR3
Radial velocity	–42.2 ± 4.2 km/s; rv_chisq_pvalue 0; rv_renormalised_gof 99.8; rv_amplitude_robust 49.8 km/s (27 epochs); vbroad 25.9 km/s	Gaia DR3
Companion size	Grazing eclipse: k = 0.34 (95%: 0.09–0.75), so R_2 > 1.7 R_{Jup} (97.5%). A non-grazing solution exists only with a circular orbit around the FLAME star, taking that density as a prior: k = 0.071, b = 0.93, ≈ 1.4 R_{Jup}. It is disfavoured by Δ(–2 ln posterior) = 32	this work

Vetting

Check	Result
Depth consistency	S7/S33 scale factors 0.997 ± 0.015 / 1.003 ± 0.015 (χ^2 0.08/1); S108 0.89 in our photometry
Independent reductions	QLP, TESS-SPOC and SPOC 2-min depths agree within 4%; a separate detrending code gives 2.97 / 3.09 ppt
Difference images	S33: 2.9" ± 1.0" from target (SNR 15); S108: 2.2" ± 0.8" (SNR 17); S7: 6.3" ± 1.1" (0.3 px, Cycle-1 30-min FFI). Validated on TOI-5542 b and TOI-2589 b (on target within 1–3")
Aperture test (S108)	Depth 2.77 / 2.78 / 2.69 ppt for 3 / 9 / 21-pixel apertures; background flat (165.7 vs 165.8 e/s)
Centroids (SPOC)	No in-transit shift: 1.1σ and 0.8σ
Data quality	No momentum dumps, quality flags or segment edges near any eclipse

Neighbours	Brightest within 40": two G = 13.5 stars at 17" and 24". Either would need a ~25% eclipse, and the difference images do not favour them
Secondary eclipse	None at phase 0.5 (–0.05 ± 0.07 ppt). No significant dip at other phases; the largest features are ~0.2 ppt, at the noise level
Catalogues	Not in the TESS EB catalogue (Prša et al. 2022), not a TOI, no confirmed planet; not in Hawthorn et al. (2024) despite fitting its S1–13 + S27–39 duotransit pattern

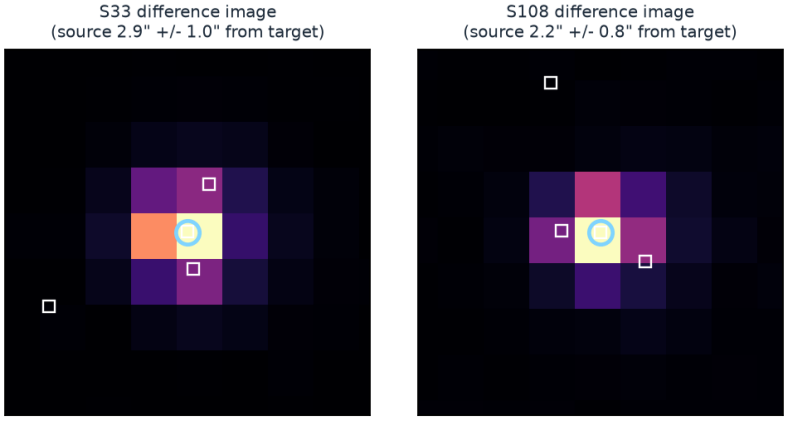


Figure 3. Difference images (out-of-eclipse minus in-eclipse) at the S33 and S108 eclipses. Blue circle: target; white squares: Gaia stars with G < 15.

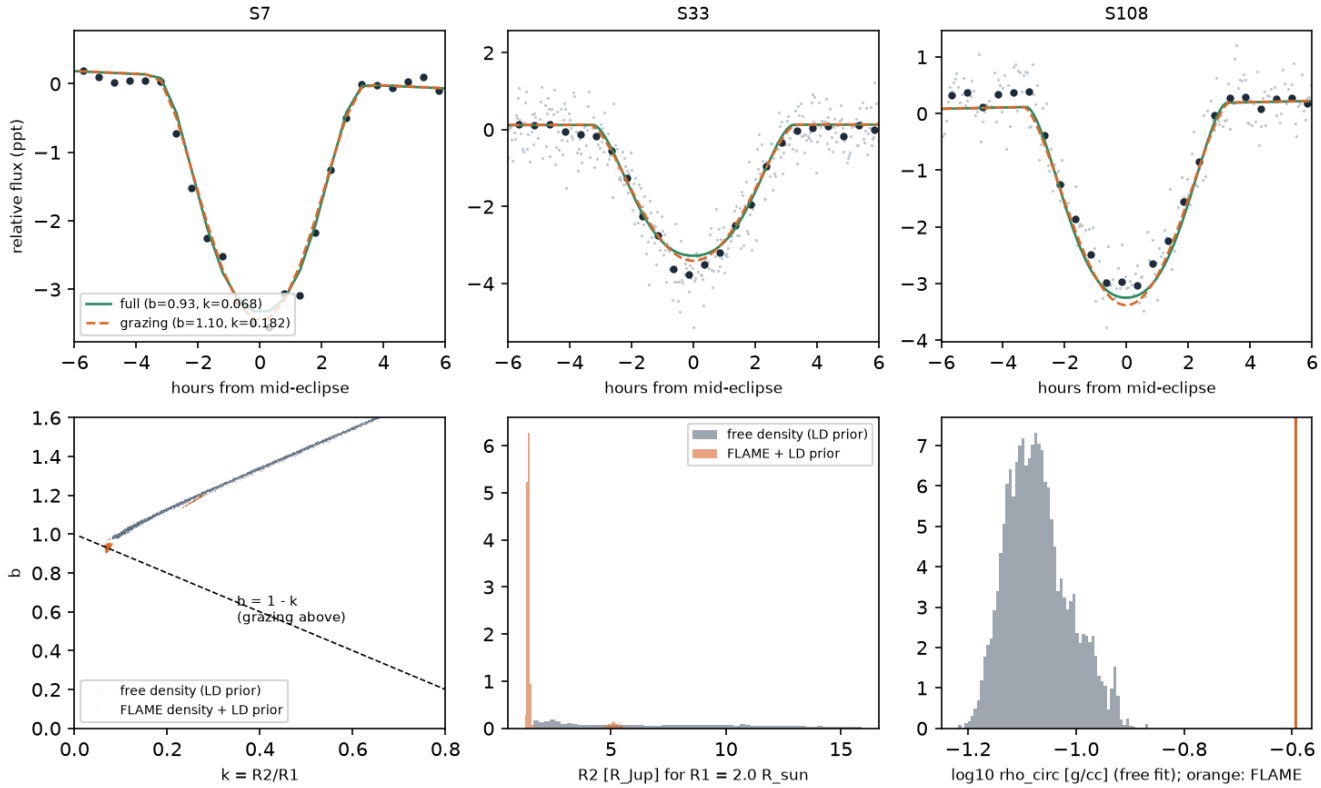


*Figure 4. Eclipse shape. **Top:** the three eclipses with the best full-transit model (green, pinned at the grazing limit b = 1 – k) and the best grazing model (orange dashed). By eye they look alike, but the likelihood favours grazing (ΔBIC ≈ 10). **Bottom** (limb darkening held to physical values): b vs k, where above the dashed line is grazing; companion radius for R_1 = 2.0 R_{sun}; and circular-orbit stellar density from the free-density fit vs FLAME.*

Method (brief)

Repeat search: repeats were found by testing a box of the catalogued depth and duration on a D/4 grid over all SPOC 2-min, TESS-SPOC and QLP light curves. Each window has its own segment-aware polynomial baseline.

Validation: run blind from the first transit only, on 26 TESS duotransit hosts with known periods, the method kept the true period every time (26/26) and recovered it uniquely for 19, with no wrong unique answers.

Alias test: aliases P = ΔT/n were tested against flat-window exclusions in all data.

Sector 108 photometry: TESScut cutout, background-subtracted aperture photometry. As a check, the same code recovers the known S33 eclipse at 3.21 ± 0.07 ppt.

Mid-times: each eclipse was fitted with a common batman transit shape and a linear ephemeris by weighted least squares. Code and intermediate files are available on request.

Suggested follow-up

(1) **RV monitoring.** A few epochs over ~40 days would show whether the Gaia RV variability follows the 37.3-d period, with a stellar-mass semi-amplitude of tens of km/s.

(1b) **Gaia DR4 (2 Dec 2026).** Phase the epoch RVs on 37.315 d. Check whether the spectrum is single- or double-lined, and look for a 37-d photocentre wobble in the epoch astrometry. This is free and decisive.

(2) **Photometry of the 2027-01-20 eclipse.** It is well placed for southern facilities, and a depth comparison in two bands tests for colour dependence.

Cycle	Mid-eclipse (UTC)	σ (min)	Alt. at Paranal	Sun alt.
76	2026-11-06 12:04	3.4	41°	29°
77	2026-12-13 19:37	3.4	-44°	49°
78	2027-01-20 03:11	3.4	79°	-39°
79	2027-02-26 10:44	3.5	-35°	1°
80	2027-04-04 18:18	3.5	27°	52°
81	2027-05-12 01:52	3.5	13°	-51°

Data: NASA TESS (SPOC, TESS-SPOC, QLP via MAST; FFI cutouts via TESScut) and ESA Gaia DR3. This analysis was carried out with substantial assistance from an AI system (Anthropic's Claude); the author has checked the results against the saved data and figures.